PTUM-F3: Proxy-Fluid vs Free-Streaming Dark Radiation

The fluid nature ($w = 1/3$, $c_s^2 = 1/3$, $\sigma = 0$) removes the neutrino-induced CMB acoustic phase shift.

A. THEORETICAL IDENTIFICATION

Property	Proxy-Fluid Dark Radiation (PTUM)	Free-Streaming Dark Radiation (e.g. ν-like)
Equation of state (w)	1/3	1/3
Sound speed (c_s^2)	1/3	1/3
Anisotropic stress (σ)	$\sigma = 0$ (exact)	$\sigma > 0$ (non-zero)
Perturbation equation	Single-fluid wave equation (no shear)	Boltzmann hierarchy with shear
Neutrino-like CMB acoustic phase shift	Absent ($\Delta\phi \approx 0$)	Present ($\Delta\phi > 0$)
Physical origin	Collective, interacting proxy-fluid sector	Collisionless free-streaming particles
Impact on CMB	No suppression & no phase shift of peaks	Peak suppression + phase shift
Effective parameter	ΔN_{fluid}	ΔN_{eff} (free-streaming)

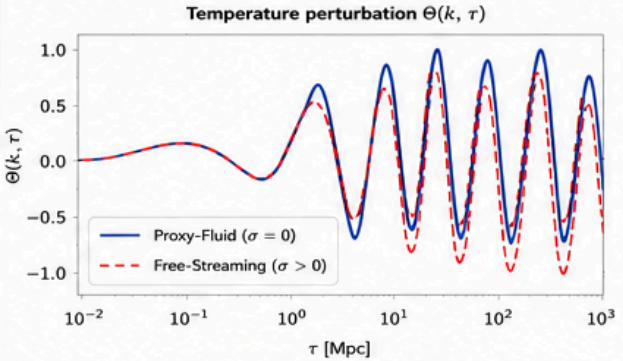
Key PTUM Prediction

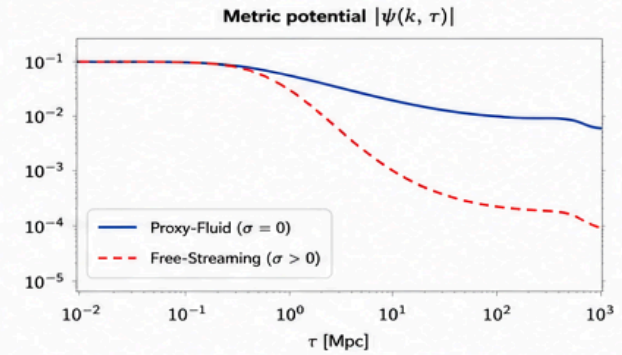
Extra relativistic energy density can exist without the neutrino-like CMB acoustic phase shift.

This is the principal falsifiable signature of PTUM.

B. PERTURBATIONS: EVOLUTION COMPARISON

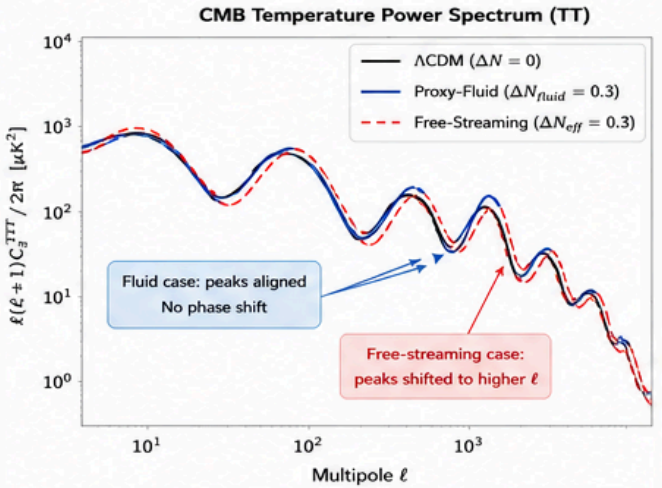
Synchronous gauge – Fourier mode ($k = 0.1\ h\ \mathrm{Mpc}^{-1}$)

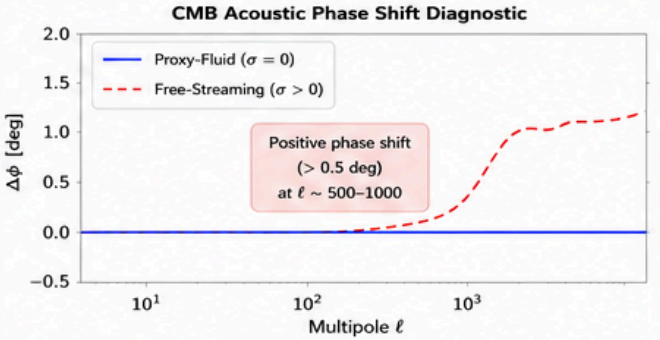




Free-streaming species ($\sigma > 0$) generate anisotropic stress that sources metric decay inside the horizon, altering the gravitational driving of acoustic oscillations. The **proxy-fluid ($\sigma = 0$) does not.**

C. CMB OBSERVABLES & PHASE SHIFT





Phase shift $\Delta\phi$ is computed from the displacement of acoustic peaks relative to ΛCDM. The **proxy-fluid case is consistent with $\Delta\phi \approx 0$.**

Physical Meaning

Anisotropic stress from free-streaming species shifts the zero-points of acoustic oscillations to higher multipoles.

A perfect fluid with $c_s^2 = 1/3$ and $\sigma = 0$ preserves the standard phase.

Observable Test

Detecting extra relativistic energy density without a corresponding phase shift would favor PTUM-like fluid dark radiation.

IMPLICATIONS

- PTUM allows $\Delta N_{fluid} > 0$ without neutrino-like acoustic phase shift.
- This separates PTUM from all free-streaming dark radiation models.
- Future CMB data can decisively test the fluid vs free-streaming nature of dark radiation.

OBSERVATIONAL TESTS

- Planck (TT, TE, EE) – overall constraints on ΔN_{fluid}
- CMB-S4, LiteBIRD – improved phase shift sensitivity
- BAO, BBN, H_0 – joint consistency checks
- Gravitational lensing & ISW – additional cross-checks

NEXT STEPS

1. Refine constraints on ΔN_{fluid} and $\Delta\phi$ with latest data.
2. Explore parameter space: ΔN_{fluid}, w, c_s^2, interactions.
3. Model-building: microphysical realization of the proxy-fluid sector.
4. Predict impacts on polarization (TE, EE) and lensing spectra.

LEGEND / PARAMETERS

w	Equation of state parameter
c_s^2	Physical sound speed squared
σ	Anisotropic stress
ΔN_{fluid}	Effective extra radiation (fluid-like)
ΔN_{eff}	Effective extra radiation (free-streaming)
$\Delta\phi$	Acoustic phase shift (degrees)
ℓ	CMB multipole moment

COLOR CODE

- Proxy-Fluid ($\sigma = 0$)
- Free-Streaming ($\sigma > 0$)
- ΛCDM ($\Delta N = 0$)

Figure PTUM-F3. Comparison between proxy-fluid (PTUM) and free-streaming dark radiation. The proxy-fluid component ($w = 1/3$, $c_s^2 = 1/3$, $\sigma = 0$) does not produce the neutrino-like CMB acoustic phase shift, providing a distinctive and falsifiable prediction of PTUM.